## 5.9. The Frobenius formula for the character of an induced representation

Let us now compute the character $\chi$ of $\mathrm{Ind}_H^G V$ when $(G:H) < \infty$. In each right coset $\sigma \in H\backslash G$, choose a representative $x_\sigma$.

**Theorem 5.9.1.** *One has*

$$\chi(g) = \sum_{\sigma \in H\backslash G : x_\sigma g x_\sigma^{-1} \in H} \chi_V(x_\sigma g x_\sigma^{-1}).$$

This formula is called the Frobenius formula.

**Remark 5.9.2.** If the characteristic of the ground field $k$ is relatively prime to $|H|$, then this formula can be written as

$$\chi(g) = \frac{1}{|H|} \sum_{x \in G : x g x^{-1} \in H} \chi_V(x g x^{-1}).$$

**Proof.** For a right $H$-coset $\sigma$ of $G$, let us define

$$V_\sigma = \{f \in \mathrm{Ind}_H^G V | f(g) = 0 \ \forall g \ \notin \ \sigma\}.$$

Then one has

$$\mathrm{Ind}_H^G V = \bigoplus_\sigma V_\sigma,$$

and so

$$\chi(g) = \sum_\sigma \chi_\sigma(g),$$

where $\chi_\sigma(g)$ is the trace of the diagonal block of $\rho(g)$ corresponding to $V_\sigma$.

Since $g(\sigma) = \sigma g$ is a right $H$-coset for any right $H$-coset $\sigma$, $\chi_\sigma(g) = 0$ if $\sigma \neq \sigma g$.

Now assume that $\sigma = \sigma g$. Then $x_\sigma g = h x_\sigma$ where $h = x_\sigma g x_\sigma^{-1} \in H$. Consider the map $\alpha : V_\sigma \to V$ defined by $\alpha(f) = f(x_\sigma)$. Since $f \in V_\sigma$ is uniquely determined by $f(x_\sigma)$, $\alpha$ is an isomorphism. We have

$$\alpha(gf) = g(f)(x_\sigma) = f(x_\sigma g) = f(h x_\sigma) = \rho_V(h) f(x_\sigma) = h\alpha(f),$$

and $gf = \alpha^{-1} h \alpha(f)$. This means that $\chi_\sigma(g) = \chi_V(h)$. Therefore

$$\chi(g) = \sum_{\sigma \in H\backslash G, \sigma g = \sigma} \chi_V(x_\sigma g x_\sigma^{-1}).$$